



SAEISS Electric Four Wheeler Design Challenge 2025



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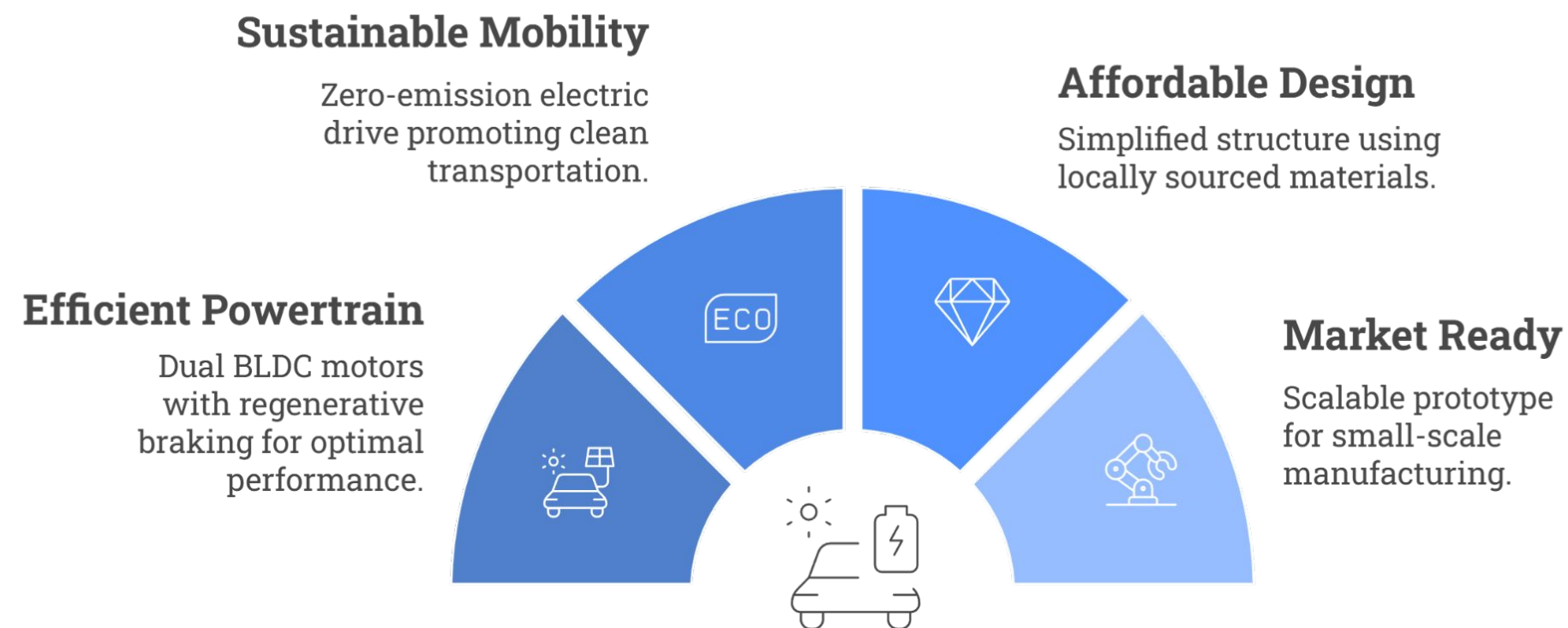
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Project Objective

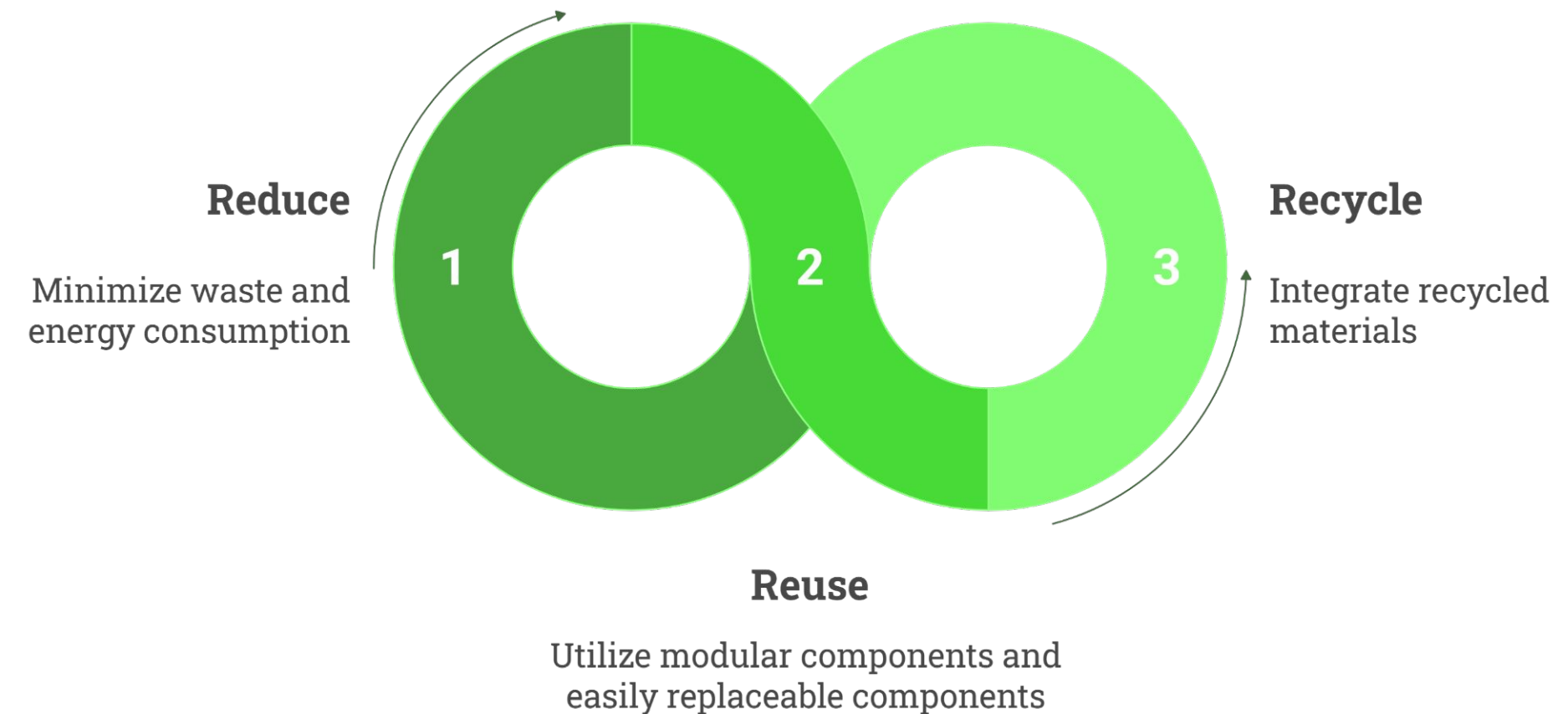
Objective

- To design and develop a compact, efficient, and affordable electric four-wheeler built for urban and campus mobility
- Focused on sustainability, simplicity, and manufacturability.
- The goal is to develop a vehicle that's easy to manufacture, energy-efficient, and reliable for real-world short-distance commuting.

Core Objectives of Electric Vehicle Design



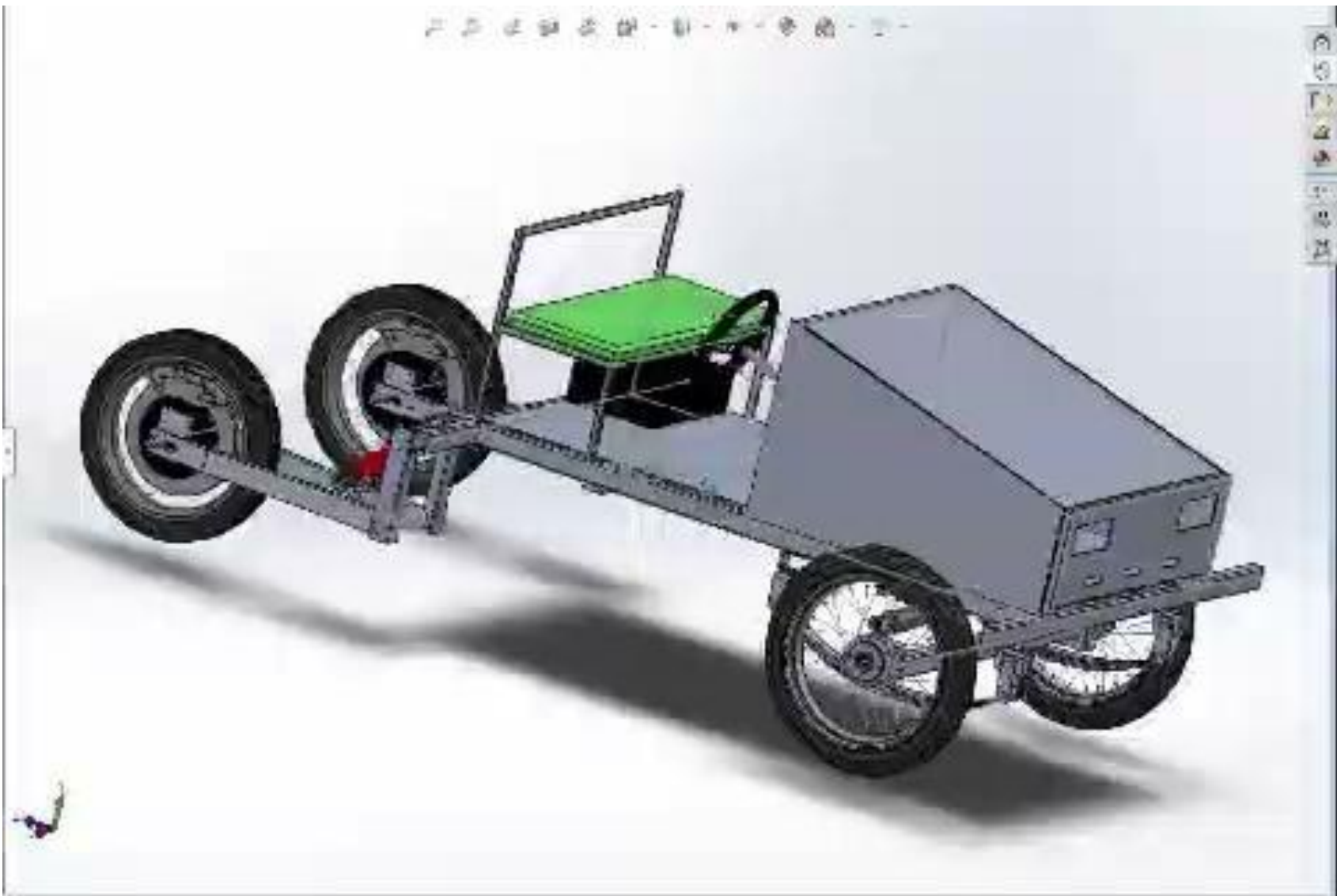
3R Approach



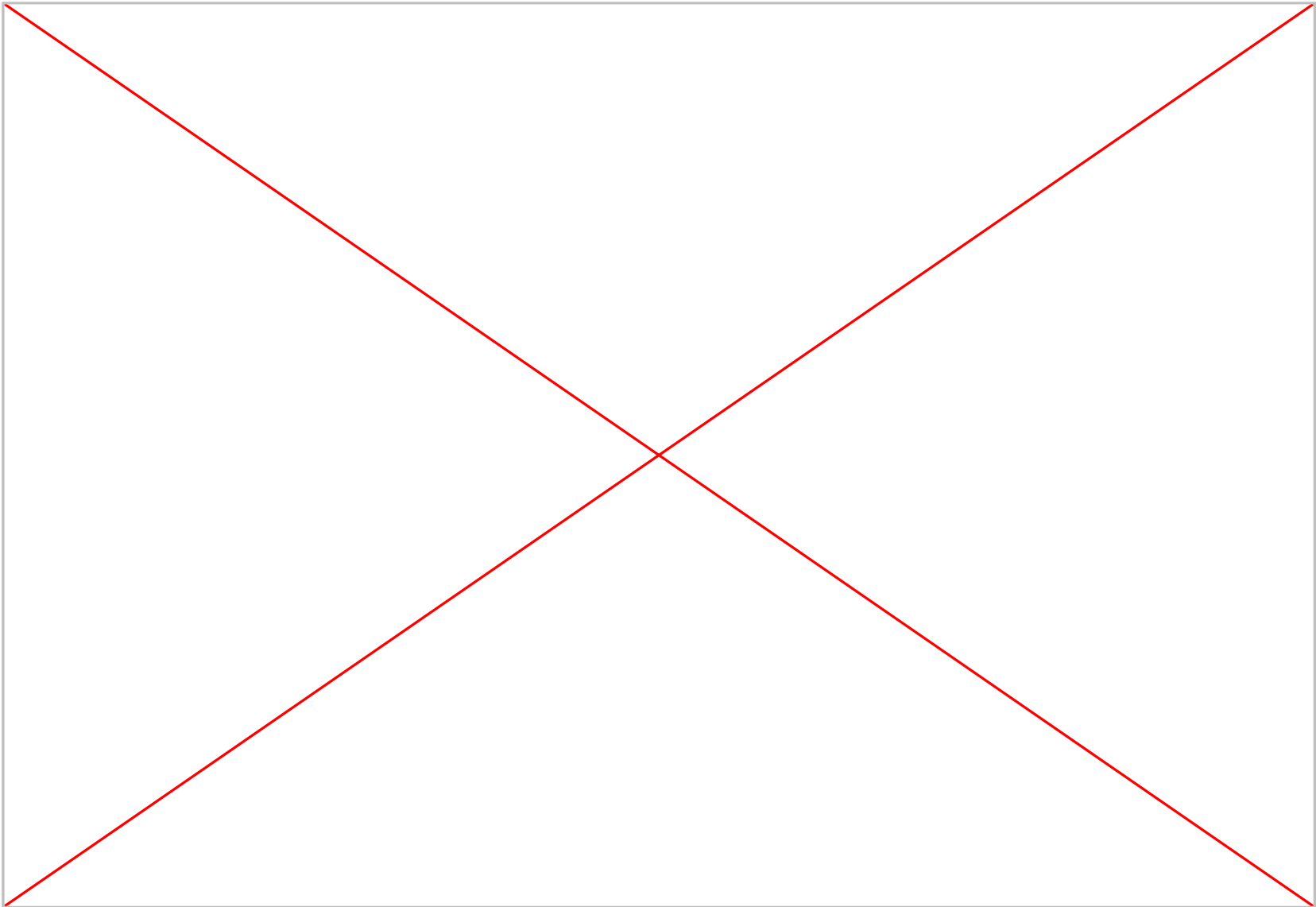
Design And Simulations



Overall Vehicle's CAD Design



Simulations in Ansys



TOTAL DEFORMATION	
LOAD	DEFORMATION
1V(1000N)	0.00011
2V(2000N)	0.00023
3V(3000N)	0.000356

Motor Speed Control

- Simulated in MATLAB/Simulink using dual 2 kW BLDC motors
- Sine-wave controller with closed-loop PI control
- Achieved smooth acceleration and stable speed response.

Charging Control

- 48 V 60 Ah Li-ion pack with Smart BMS
- Simulated charging cycle with SoC, voltage & temp monitoring
- Verified safe cutoff and regen energy recovery

Matlab/Simulink simulation

MOTOR SPEED CONTROL SIMULATION

Overall Vehicle Specifications



CHASSIS & SUSPENSION

- Ladder type- Chassis material- IS 4923
- Frame material IS 18753
- Front Suspension - leaf spring
- Rear Suspension - Independent Open Coil Suspension

PERFORMANCE

- Top Speed : 35 km/h .
- Range : ~50 km per full charge
- Initial Torque : 38.26 Nm
- Gradeability : 20

PROPULSION SYSTEM

- Battery : 48V-60Ah - li-ion (NMC)
- Using dual motor for rear drive
- Motor : 2KW BLDC HUB Motor
- Controller : Sine Wave (42-72V, 30-40 Amps)

BREAKING SYSTEM

- Using both hydraulic disk brake and regenerative-braking.
- Using Disk brake for rear
- Break oil master cylinder : Tandam
- Break oil : DOT-4
- Brake Caliper : dual piston



Motor, Battery and Controller



MOTOR

- Equipped with dual 2 kW **BLDC hub motors** mounted on the rear wheels.
- A **direct-drive setup** no chains or gears, which **reduces energy loss** and maintenance.
- Delivers instant torque and a smooth.
- Both motors support **regenerative braking**, improving efficiency and range.

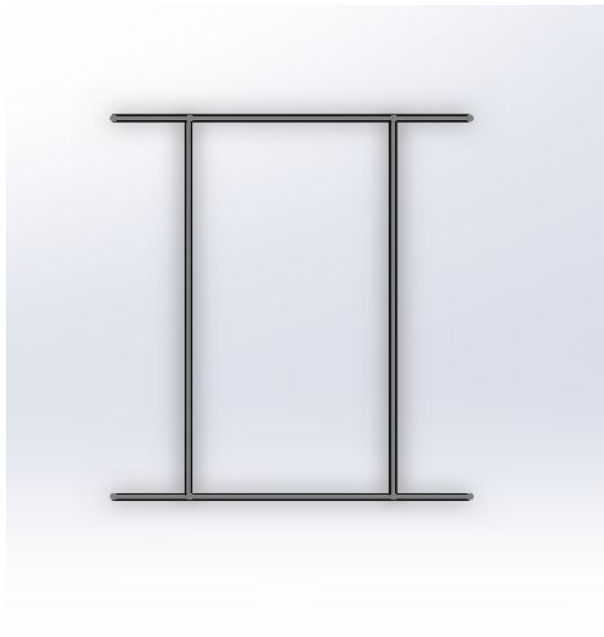
BATTERY

- Powered by a 48V, 60Ah lithium-ion (NMC - nickel manganese cobalt) smart battery pack.
- Integrated Bluetooth-enabled BMS tracks battery health, voltage, and temperature in real time.
- Ensures safe operation and longer battery life.
- Designed for reliability and consistent power delivery.

CONTROLLER

- Sine Wave Controller (48/60V, 30–40 A)
- Delivers smooth torque response and motor synchronization
- Controls regen braking & dual-motor efficiency
- Built-in over-current, over-voltage, and thermal protection

Chassis, Suspensions, Breaking and Steering



Chassis

- **Type:** Ladder Frame Structure
- **Material:** Mild Steel (IS 4923) – high strength, square tubular section.
- **Design Focus:** Optimized rigidity with reduced weight.
- **Manufacturing:** ARC welded frame for easy fabrication & durability.



Suspensions

- **Front:** Leaf Spring – simple, rugged, and cost-effective
- **Rear:** Independent Coil Spring – improves ride comfort and traction
- **Combined Effect:** Balanced stability and smooth handling



Breaking

- **Type:** Dual Hydraulic Disc Brakes (Rear Only)
- **Support:** Regenerative Braking via Sine Wave Controller
- **Advantage:** Better stopping power, improved energy efficiency



Steering

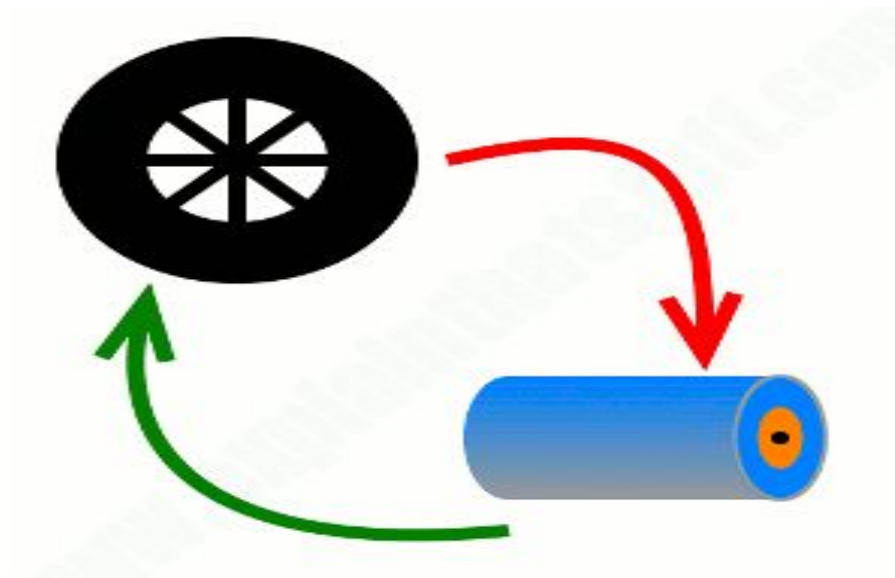
- **Type:** Manual Rack & Pinion (Maruti Omni System)
- **Feature:** Compact setup with low steering effort
- **Turning Radius:** ~3.8 m – ideal for tight spaces and campus roads

Innovation



Rear Swing Arm Suspension:

- Provides enhanced ride stability and comfort on uneven surfaces.
- Swing-arm geometry improves traction and handling during turns.
- Built with mild steel tubing, ensuring strength, durability, and ease of manufacturing.



Coasting Regen:

- Simultaneous regenerative and mechanical braking for efficient control.
- Converts kinetic energy into electrical power, recharging the battery during deceleration.
- Enhances range by 8–10% and reduces mechanical wear and energy losses.



Direct-Drive Hub Motor:

- Rear-mounted 2kW BLDC hub motors eliminate transmission losses.
- Provides high torque, silent operation, and instant acceleration.
- Compact and low-maintenance design simplifies powertrain architecture.

"Innovation that drives efficiency, comfort, and sustainability!"

USP and Business Scope



Business Scope

Urban Delivery Advantage: Engineered for reliable, all-weather delivery operations with a silent electric drivetrain that ensures cost-effective, uninterrupted, and neighborhood-friendly service offering a smarter, cleaner alternative to conventional two-wheelers.

University, Industrial & Institutional Mobility: Designed for safe, low-speed, and eco-friendly transportation within campuses, resorts, and industrial zones offering quiet, efficient, and sustainable movement for passengers and utility operations



USP

- **Dual Hub Drive:** Simple, reliable, and low-maintenance drivetrain.
- **Smart BMS:** Bluetooth-enabled Li-ion battery for real-time monitoring.
- **Dual Braking:** Regenerative + hydraulic system for safety & efficiency.
- **Modular Design:** Easy to build, assemble, and scale.
- **Eco & Cost Efficient:** Lightweight, energy-saving, and affordable mobility.

Demo Video And Survey



Demo Highlights:

- Successful on-road demonstration of vehicle performance.
- Smooth acceleration, stable handling, and effective braking.
- Regenerative braking and dual-motor synchronization validated in real time.
- Battery and control system performed efficiently under load conditions.

Public Survey Insights:

- Conducted across students, staff, and local commuters.
- 92% rated the ride comfort and noise-free operation as excellent.
- 85% appreciated the eco-friendly and cost-efficient design.
- Strong interest shown for campus mobility and last-mile delivery use.

